

YEAR 10 SCIENCE (ADJUSTED)
PHYSICAL SCIENCE
TOPIC TEST

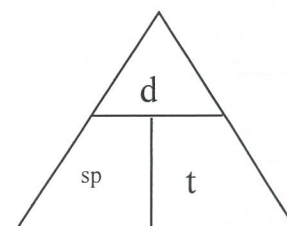
Name: SOLUTIONSMark: 18**Part 1: Multiple Choice**

Write your choice for each question in the table below.

1. Speed is a measure of the rate:
- (a) at which distance is covered.
 - (b) at which speed increases.
 - (c) of change in force.
 - (d) of change in acceleration.
2. Belinda walks a total distance of 1200 metres to school each morning. If this trip takes her 800 seconds, her average speed would be:
- (a) 1.5 m/s.
 - (b) 0.67 m/s.
 - (c) 0 m/s.
 - (d) 960,000 m/s.

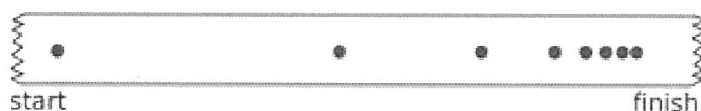
MULTIPLE CHOICE ANSWERS	
1	A
2	A
3	B
4	D
5	B
6	D

3. A car is driven along a highway at an average speed of 90 km/h for 2 hours without a rest stop. What distance has it covered?
- (a) 45 km
 - (b) 180 km
 - (c) 360 km
 - (d) Not enough information is given to calculate.



4. An object moves with a speed of 5 m/s for 15 seconds. What distance has it travelled?
- (a) 3 m
 - (b) 10 m
 - (c) 60 m
 - (d) 75 m
5. A car travels 400 km on a highway at a speed of 80 km/h. What time does the trip take?
- (a) 0.5 hours
 - (b) 5 hours
 - (c) 20 hours
 - (d) 50 hours

6. The sample of ticker tape below shows:



- (a) a uniformly accelerating vehicle.
- (b) a vehicle travelling at a constant velocity.
- (c) a stationary vehicle.
- (d) a uniformly decelerating vehicle.

Part 2: Short Answer

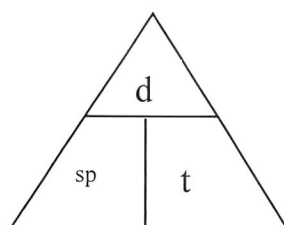
1. Describe or define the following terms. (2 marks)

(a) distance: *how far an object moves in any direction.* (1)

(b) displacement: *how far an object moves in a straight line in a particular direction.* (1)

2. Complete the following table on **speed** calculations. Make sure you include the appropriate units for each quantity.

(3 marks)



Distance travelled	Time taken	Average speed
490 m	35 s	14 m/s
276 m	12 s	23 m/s
121 m	11 s	11 m/s

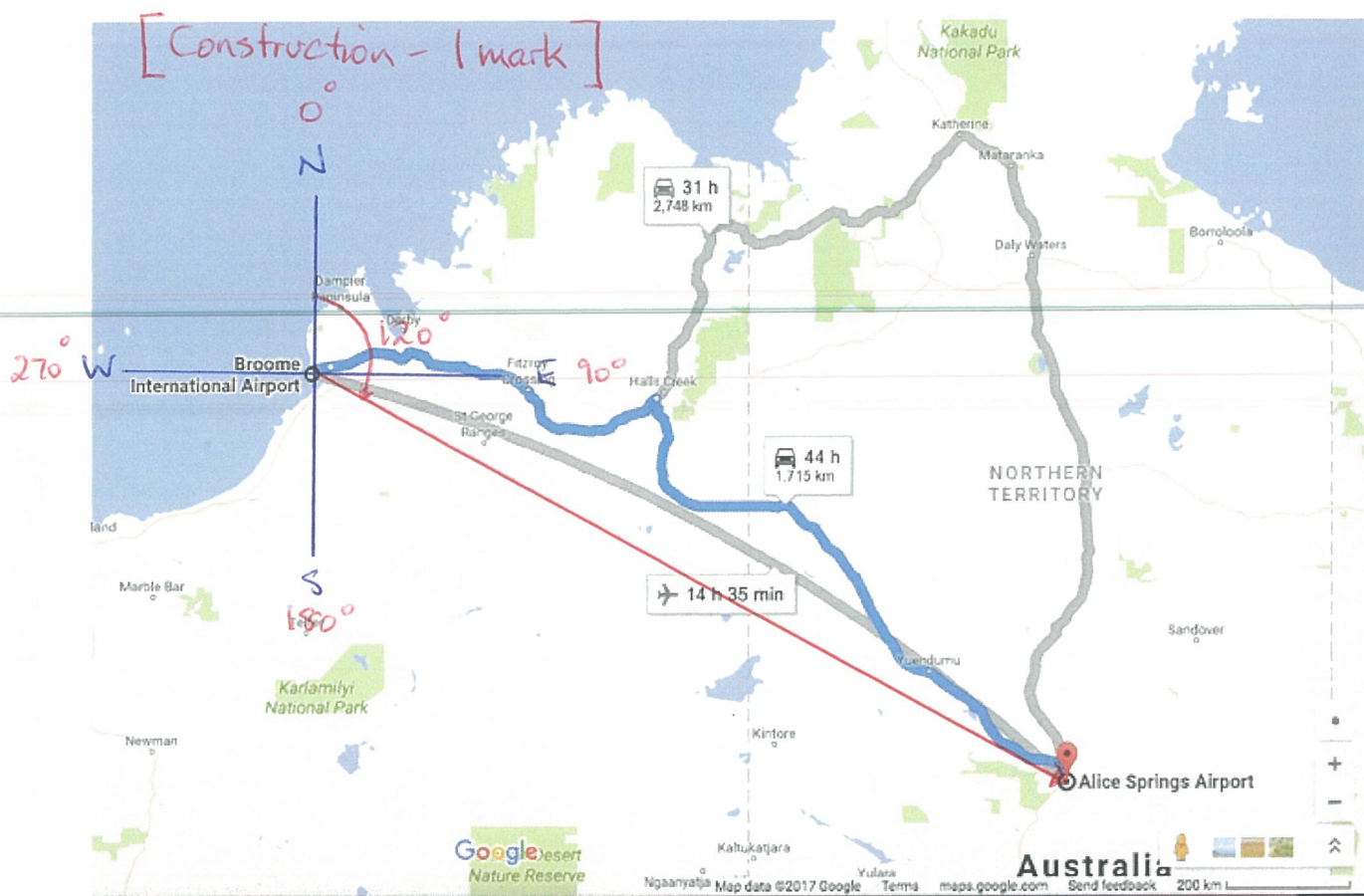
3. The map below shows Broome and Alice Springs. Use the map (look at it carefully) and its scale to work out the following.

- (a) The distance between Broome Airport and Alice Springs airport on the road. (1 mark)

1715 km (1)

- (b) The displacement between the two airports - start from Broome.
(HINT: Don't forget to include a bearing.) (3 marks)

(1) 1400 km bearing 120°
(± 100 km) ($\pm 10^\circ$)



4. Decide whether the following are true (T) or false (F).

(3 marks)

- Average speed = distance $\div$ time.
- A swimmer who does 2 laps of a 50 m pool travels a distance of 100 m.
- The relationship between time, distance and speed is $t = \frac{sp}{d}$.

T

T

F

[1 mark each]

